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Designing an External Storage Device

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Prof Michael Collins
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Dear Professor Collins,

This report, entitled Designing an External Storage Device, was written to fulfil my 3B work term report requirements. This is my 3rd work report submission.

The idea for this project came to me after the sudden failure of my external hard drive. I had always carried an external hard drive on me to use for transporting large files and backing up my data. The drive was a Seagate mechanical hard drive which had worked with no issues for years prior. After it was clear I needed to re-think my future storage method.

The external drives being offered off the shelf were expensive and lacked all the features I wanted. To find an external drive which would match my strict criteria I would have to build my own.

This report was written entirely by me, and has not received any previous academic credit at this or any other institution.

Best Regards,

Kwaku Richter
ID: 20851201
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Table of Contents

List of Figures	ii
List of Tables	iii
Summary	iv
1.0 Introduction.....	1
1.1 Hard Drive Construction and Operation	1
1.2 Flaws	2
1.3 Read and Write Speed.....	2
1.4 Objective	3
2.0 Problem Definition.....	3
2.3 Constraints and Criteria	3
3.0 Design and Analysis	5
3.1 Hardware Selection.....	5
3.1.1 Storage Medium.....	5
3.1.2 SSD Type/Model	6
3.1.3 Connectivity	7
3.1.4 Controller	7
3.1.5 Enclosure.....	9
3.2 Potential Modifications	10
3.2.1 Background Theory.....	11
3.2.2 Geometry.....	14
3.2.3 Material	16
3.2.4 CFD Simulations.....	17
3.2.3 Results.....	19
3.3 Remaining Challenges	23
4.0 Conclusions.....	23
4.1 Conclusions.....	23
4.2 Next Steps / Recommendations	24
References.....	25

List of Figures

Figure 1: Diagram of modern hard drive [3].....	1
Figure 2 – Typical SATA SSD Construction [6]	6
Figure 3 - Sequential Write Speeds of 980 Pro SSD [14]	8
Figure 4 - Comparison of Sustained Write Speeds [14]	9
Figure 5 - Trebleet 2 in 1 TB3 Enclosure [17].....	10
Figure 6 - Cross sectional Diagram of Enclosure	11
Figure 7 - Expected Heat Transfer	12
Figure 8 - Thermal Resistance Network	13
Figure 9 - CAD Model of Trebleet Enclosure	15
Figure 10 - Split CAD Model of Trebleet Enclosure	16
Figure 11 - Proposed Fin Designs (left - plate fin, middle - cylindrical fin, right - pin fin)	16
Figure 12 - Exterior Fluid Boundary Rule of Thumb [21].....	17
Figure 13 - Initial Air Speed Domain	18
Figure 14 – Average Temperature of SSD.....	20
Figure 15 - Rate of Average SSD Temperature Change	21
Figure 16 - Temperature of Aluminum (top) and Copper (bottom) at 10, 30, and 50 s (left to right)	22
Figure 17 - Render of Final Enclosure Design	23

List of Tables

Table 1 - Connectivity Options 7

Summary

Following the fatal malfunction of a commercial external hard drive resulting in the complete loss of data, a need was identified to design a superior replacement. Above all this replacement device had to be more robust and reliable to avoid a failure. The device had to support all speed/connectivity standards, store at least 1 TB of data, be easily portable, and function without internet connectivity. Maximizing transfer speeds and cooling performance while minimizing costs was also to be considered.

Given these requirements, a suitable external storage device was designed with a combination of both commercially available and custom parts. SSD's were chosen instead of the legacy mechanical hard drives to improve reliability and speed. Specifically, Samsung's 980 Pro 1 TB SSD was selected for its impressive price to performance ratio. Trebleet's 2 in 1 TB enclosure was paired with the 980 Pro due to its combination of connectivity options, volumetric footprint, cost, and availability.

To further improve the transfer speed and cooling performance of the replacement device, natural convection modelling was conducted with CFD. Deriving a relationship which describes the rate of heat transfer from the SSD to the ambient air indicated modifications which would theoretically increase the cooling performance. The first modification was to increase the exterior surface area by adding cooling fins. The enclosure could also be manufactured with copper instead of aluminum to increase its thermal conductivity. Simulations then verified these posits by providing sufficient data to compare performance of the proposed modifications to the factory design.

The next step is to use these findings to assemble the necessary parts to build the replacement external storage device. Once constructed, the new enclosure can be tested to ensure it satisfies all constraints and criteria in practice

1.0 Introduction

Given the vast amount of digital data that most individuals interact with on a daily basis, the equally vast demand for suitable methods to store that data comes with no surprise. Beginning in the 1950's, the first commercially available forms of digital computer storage were in the form of 50 24-inch platters which contained a total of five million 6-bit characters, or approximately 3.75 megabytes [1]. These were massive units which weighed in excess of one ton and had to be moved with forklifts [1].

Over the past 70 years the capacity of these drives has grown exponentially while the physical size continually decreased. Today, modern magnetic hard drives can store up to 16 terabytes of data, over 4 million times the storage of the legacy hardware, in a 3.5-inch enclosure [2].

Although the capacity of these modern drives is impressive, there are several inherent disadvantages which become severely limiting in certain use cases.

1.1 Hard Drive Construction and Operation

Magnetic hard drives generally consist of a spindle that holds a series of flat circular disks (platters) which are coated in a thin layer of magnetic material. Beside the spindle and platters is the actuator assembly which houses the actuator arm and heads (Figure 1).

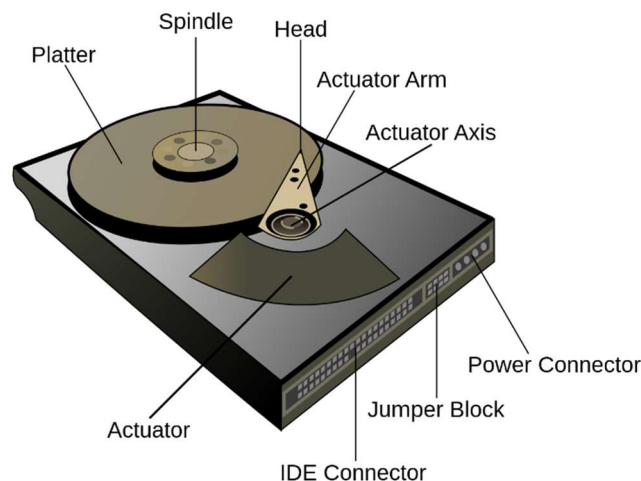


Figure 1: Diagram of modern hard drive [3]

The platters are spun by a motor attached to the spindle at speeds of 5400-7200 rpm. Data is stored in grain domains in a layer of magnetic material on the platters. Within any given grain domain, several individual grains are magnetized all in the same direction. Together these grains represent one node of data. During read operations the read head analyzes the patterns in these grain boundaries and outputs a stream of electrical signals which represent binary data. Similarly, during write operations the write head rearranges the respective magnetic directions of the grain domains according to the input of binary signals [3].

1.2 Flaws

In static operations that require plentiful cheap storage magnetic hard drives still offer many benefits over other technologies. However, when movement is introduced, the reliability of magnetic hard drives decreases rapidly [3]. Intrinsically, because hard drives rely on fragile spinning disks as well as multiple other moving parts to function, any movement should be kept to a minimum.

1.3 Read and Write Speed

Another critical factor which affects the usability of a storage device is the transfer speeds that device can achieve. There are generally three possible bottlenecks in most applications. The first and most common bottlenecks are the maximum read and write speeds of the storage device. In the case of magnetic hard drives for example, because the design requires moving parts to read and write data there is a practical limit to the transfer speeds. The fastest hard drive currently on the market, which spins its platters at 7200 rpm, rivals only the slowest solid state drives (SSD).

The next bottleneck is the connectivity technology. The majority of external storage devices communicate via USB (Universal Serial Bus). Beginning in the 1990's, USB has consistently iterated and expanded to become the standard of choice in connecting peripheral devices to computers [4]. From a maximum speed of 12 MB/s (megabytes per second) with USB 1.0 to 40 GB/s (gigabytes per second) with USB 4.0, the variability of possible transfer speeds using USB is immense. Given two different USB standards on either end of the transfer (external storage device or computer) the transfer speed will drop to the lowest common denominator.

The final bottleneck is a result of the computer in which the data is being copied to or from. There are various parts that work in conjunction to transfer data to and from a computer such as the CPU (Central Processing Unit) the system memory RAM (Random Access Memory) et cetera. Generally, the system

storage is responsible for bottlenecks in modern computers. The quality and speed of the system storage can vary greatly.

1.4 Objective

After 5 years of operation the authors Seagate external hard drive failed and along with it all the data it contained was lost. The need for a replacement external storage solution is clear.

The objective of this report is to design an external storage device which will match or exceed those found on the market today while keeping costs to a minimum. The device must be rugged, reliable, compact, fast, and it must remain relatively cool during normal operation.

2.0 Problem Definition

Following the malfunction of the legacy mechanical external hard drive, there is a need to replace it with an external storage device which will exceed the old one. The new device must meet the constraints and criteria listed below. These requirements are intended to exceed those of the legacy drive as well as including additional modern developments.

2.3 Constraints and Criteria

The chosen design must meet the following criteria:

1. Must be rugged, able to withstand normal operation.
2. Must support the slowest to fastest available transfer speed depending on the connected device.
3. Must store a minimum of 1tb.
4. Must be compact, able to easily fit in backpack
5. Must be operational without internet connection.

The chosen design must also adhere to the following constraints:

1. Should maximize read/write speeds.
2. Should minimize costs.
3. Should maximize heat dissipation efficacy.

3.0 Design and Analysis

3.1 Hardware Selection

3.1.1 Storage Medium

The first and most important criterion to be satisfied is the increased ruggedness and overall durability of the new storage device. There are two technologies to choose from based on this requirement; a cloud storage service and SSD's.

Cloud storage services operate large server farms globally that provide fast and safe data storage. Because these server farms are located in vast warehouses in static operation, the risk of catastrophic failure of the storage medium is diminished. Many server farms also employ some form of redundancy in their design which further minimizes the possibility of lost data. Google claims to “distribute all data — including [their] own — across many computers in different locations. [They] then chunk and replicate the data over multiple systems to avoid a single point of failure” [5].

However, these cloud storage services fall short in their required internet connection. Requiring an internet connection to access any data stored limits the practical applications. Not only will the service stop working in areas without network access, the speeds of data transfers depend on the user's ISP (Internet Service Provider) connection speed.

The second possible storage device is an SSD. Unlike mechanical hard drives SSD's do not rely on moving parts to function. Although the general operation of both designs is mostly the same, hard drives encode data via spinning platters while SSD's use flash memory. Rather than encoding each bit on a respective magnetic domain, SSD's use integrated circuits to assign the value of each bit on non-volatile semiconductive MOS (Metal-Oxide Semiconductor) gates (Figure 2). This distinction results in a storage medium which is much less prone to physical damage with higher read and write speeds.

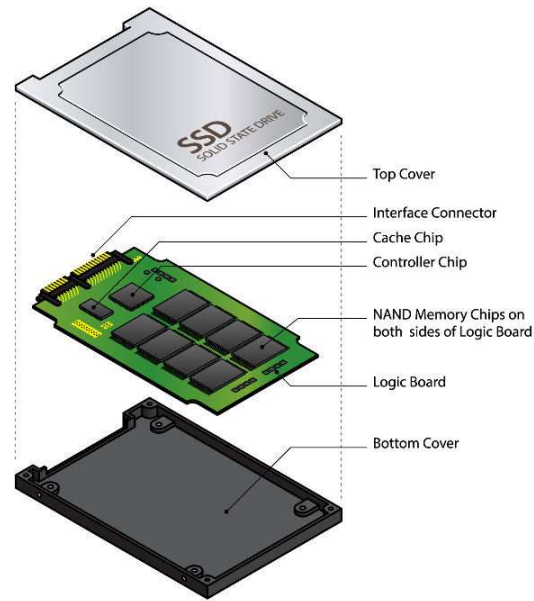


Figure 2 – Typical SATA SSD Construction [6]

3.1.2 SSD Type/Model

Modern SSD's are distinguished by their connectivity method. The two dominant designs use either SATA (Serial ATA) or NVMe (Non-Volatile Memory Express). SATA is an older bus standard which was originally designed for use with mechanical hard drives. It uses AHCI (Advanced Host Interface Controller) to communicate. SATA III is the latest iteration and offers transfer speeds up to 6 GB/s [7].

NVMe is a relatively modern invention which was designed to operate with non-volatile memory devices. This standard uses the PCIe (PCI Express) bus to communicate directly with the CPU via PCI lanes, offering significantly greater throughput. Devices using PCIe 4.0 standard speeds can transfer data with a theoretical speed of 32 GB/s using the full 16 lanes [8]. NVMe SSD's use a maximum of 4 PCI lanes and therefore have a theoretical maximum speed of 8 GB/s.

There are several competing models of PCIe 4.0 NVMe SSD's currently on the market. The fastest SSD based on sequential read and write speeds is the Western Digital SN850X, with read/write speeds up to 7300 MB/s and 6300 MB/s respectively [9]. At the time of writing, the SN850X 1 TB model is being sold for \$204.99 CAD. Optimizing for better price to performance results in the Samsung 980 Pro 1 TB SSD. The 980 Pro offers 88.4 % performance (7000 MB/s read, 5100 MB/s write), with a 36.6 % reduction in

cost (\$129.99 CAD) when compared to the SN850X [10]. Both Western Digital and Samsung offer 5-year warranties.

3.1.3 Connectivity

The next major design consideration is the connectivity method between external storage and device. The new drive is intended to be used with three devices: a custom Windows PC, a Dell XPS 15 9560, and an Apple M1 iPad Pro. These devices offer various connectivity options and speeds as shown in Table 1 below.

Table 1 - Connectivity Options

Devices	Connectivity Standards			
	USB 3.1 (10 GB/s)	USB 3.2 (20 GB/s)	TB3 (limited = 10GB/s)	TB3 (full = 40 GB/s)
Custom PC	✓	✓		
XPS 15	✓	✓	✓	
iPad Pro	✓	✓	✓	✓

It should be noted that while they may have similar transfer speeds, USB and TB3 (Thunderbolt 3) are not cross-compatible by design. TB3 is a proprietary standard developed by Intel while USB is open source.

Upon initial inspection it appears that USB 3.1 should be the connection standard of choice as the maximum transfer speed of the SSD is 7000 MB/s or 7 GB/s which is lower than the theoretical maximum of USB 3.1. Also, all three devices share this capability. However, there are other limiting factors that will be discussed next.

3.1.4 Controller

The next consideration in acquiring the necessary hardware for the storage device is the controller. When connecting peripheral devices to computers, controllers are necessary to facilitate communication. The controller allows data transfer between peripheral devices and the host device [11].

Intel currently offers a variety of controllers with differing capabilities. Intel manufactures two controllers which can provide the maximum theoretical TB3 throughput of 40 GB/s: the JL6340 and JL7440 [12].

The JL6340 however does not support additional USB 3.1 and 3.2 connectivity which is a requirement [13].

The JL7440 uniquely offers both full TB3 and USB 3.1/3.2 connectivity. It should be noted however that in practical applications the JL7440 can only deliver maximum read/write speeds of 2700 MB/s and 2000 MB/s respectively in TB3 configuration. In USB configuration this drops even further to 900 MB/s read and 500 MB/s write. It is unclear exactly why this limitation exists. One possible explanation is a result of the maximum limit of 4 PCIe 3.0 lanes in TB3 connectivity. PCIe 3.0 has a maximum theoretical transfer speed of 16 GB/s (16 000 MB/s) which equates to 1000 MB/s per lane [8]. Intel is most likely deliberately capping the NVMe throughput in order to conserve some for the other peripherals included in the TB3 standard.

At first glance the choice of the 980 Pro seems unnecessary given its 7000 MB/s maximum read speeds compared to the maximum of 2700 MB/s from JL7440. According to hardware reviewer Tom's Hardware, the 1 TB Samsung 980 Pro is able to achieve write speeds equivalent to the advertised speed of 5100 MB/s [14]. It does this by utilizing its so called 'Intelligent TurboWrite 2.0' which uses 114 GB of high-speed cache storage to temporarily speed up relatively small operations. However, once this cache has been fully saturated after around 25 seconds the speed drops dramatically to around 2000 MB/s as shown in Figure 3 below.

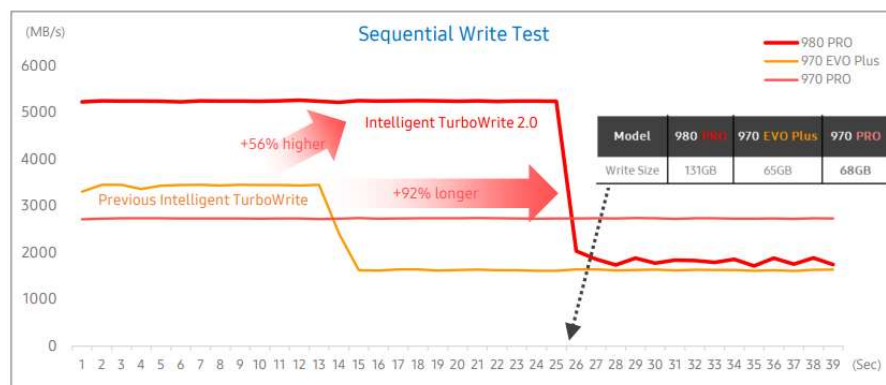


Figure 3 - Sequential Write Speeds of 980 Pro SSD [14]

This results in large file transfers between the host device and the external SSD being mostly unaffected by the bottleneck introduced by the JL7440 controller. Looking at the sustained performance of the 980 Pro versus an alternative such as the Western Digital SN550 which reports maximum read/write speeds of 2400 MB/s and 1950 MB/s respectively, the 980 Pro is shown to maintain around 2000 MB/s while the SN550 drops to 800 MB/s (Figure 4). This 60 % drop in performance only results in a 11.5 % lower price (\$114.99 CAD) at the time of writing [15].

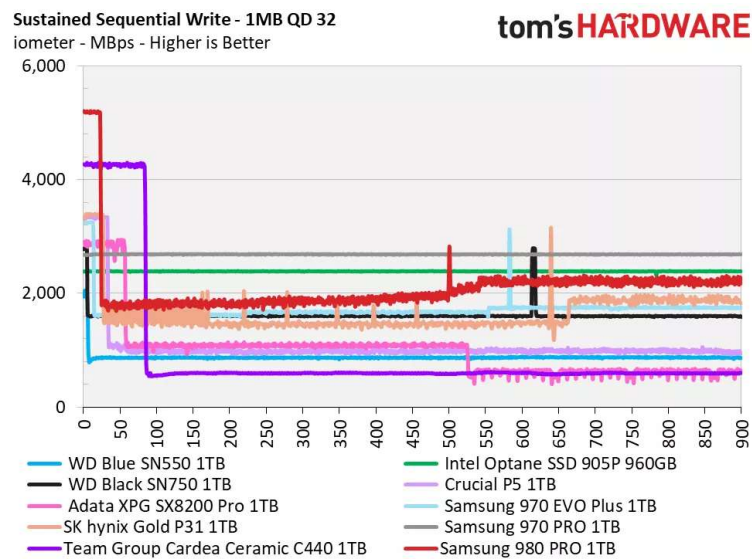


Figure 4 - Comparison of Sustained Write Speeds [14]

The exceptionally high sustained read/write speeds of the 980 Pro SSD, along with the remarkably low retail price makes it a good choice to combine with the JL7440 controller.

3.1.5 Enclosure

The final hardware acquisition necessary for the external storage device is the physical enclosure which houses the SSD. There are small number of manufacturers currently selling external NVMe SSD enclosures which incorporate the JL7440 controller. The first available option is the JEYI ThunderDock TB3. This enclosure supports USB 3.1/3.2 and full TB3 connectivity. It comes in a sleek Aluminum case (105mm x 56mm x 20mm). It also includes various additional functionality in the form of:

- 2 – TB/USB 3.1/3.2 ports (USB C connector).

- 1 – DisplayPort 1.4.
- 1 – USB 3.1 (USB A connector).

However, the enclosure is listed for \$209.99 CAD retail and is currently out of stock [16].

The other enclosure which meets the necessary requirements is the Trebleet 2 in 1 TB3 dock (Figure 5). This dock integrates a JL7440 controller with USB 3.1/3.2 and TB3 support. It also includes an additional JMicron JMS583 controller which Trebleet claims will increase the efficacy of its dedicated USB 3.1/3.2 connectivity. The enclosure similarly comes in an Aluminum case (98mm x 49mm x 14.5 mm). It is listed at \$139.99 CAD [17].



Figure 5 - Trebleet 2 in 1 TB3 Enclosure [17]

3.2 Potential Modifications

Unlike mechanical hard drives, one of the disadvantages of modern SSD's is their tendency to produce a lot of heat during heavy operation. As a result, SSD's have temperature sensors which monitor various critical points on the device. Past a certain threshold a controller will begin to limit the transfer speeds of the SSD in an attempt to reduce the temperature, also known as 'thermal throttling'. Generally, this occurs during sustained operation of the SSD where the cooling solution does not match the thermal load.

The SSD enclosure must remain small, portable and rugged. This means bulky and delicate active cooling solutions such as fans are not an option. The enclosure must therefore rely on passive natural convection to disperse the generated heat. To further increase the performance of the enclosure, modifications to the factory Trebleet shell are required.

3.2.1 Background Theory

Consider the heat diffusion equation (Equation 1) defined in 4 dimensions (x, y, z, t). A small section of the top of the enclosure is shown below (Figure 6). Assume the device has been operating for some time and has reached equilibrium (steady state) with its environment. To further simplify, consider the flow of heat from the SSD to the exterior in 1 dimension. Assume also that the density and thermal conductivity remain constant throughout (Equation 2).

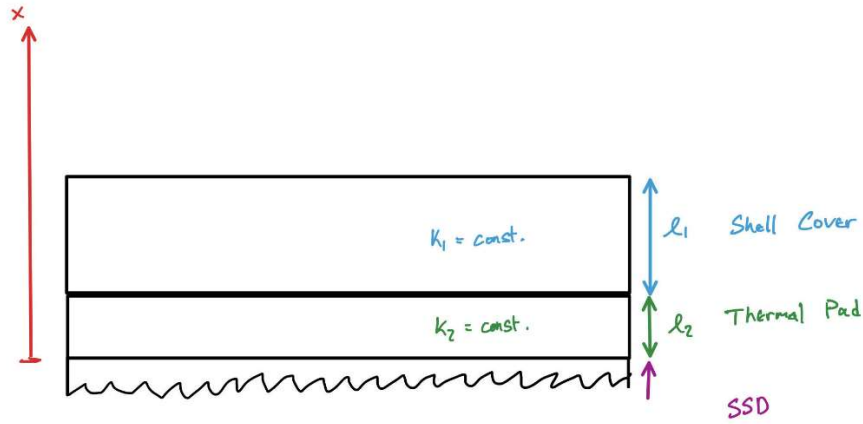


Figure 6 - Cross sectional Diagram of Enclosure

$$\vec{\nabla} \cdot (k \vec{\nabla} T) + q_g''' = \rho c_p \frac{\partial T}{\partial t}$$

Equation 1 - 4D Heat Diffusion Equation [18]

Where:

- k = thermal conductivity $[\frac{W}{mK}]$
- q_g''' = rate of heat generation $[\frac{W}{m^3}]$
- ρ = density $[\frac{kg}{m^3}]$

- c_p = specific heat [$\frac{J}{kg \cdot K}$]
- T = temperature [K]
- t = time [s]

$$\frac{\partial}{\partial t} \left(k \frac{\partial T}{\partial x} \right) + \cancel{q_g''' } = \cancel{\rho c_p} \frac{\partial T}{\partial t}$$

$$\frac{d^2 T}{dx^2} = 0$$

Equation 2 - 1D, Steady State, Without Heat Generation

Where:

- x = distance from SSD [m]

The second law of thermodynamics states that the entropy of an isolated system increases. Plainly, heat flows from areas of higher temperature to those of lower temperatures. In this example, heat is expected to conduct from the top surface of the SSD through the thermal pad and top cover. Convection through the ambient air is expected from the exterior surface of the enclosure (Figure 7).

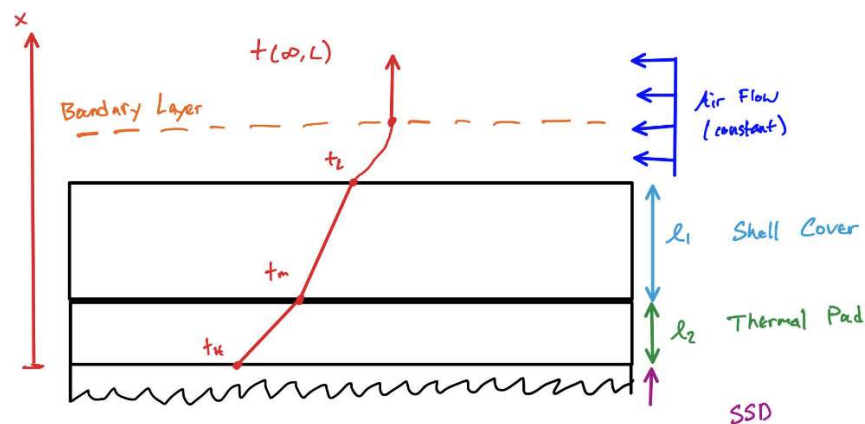


Figure 7 - Expected Heat Transfer

Applying the thermal resistance network method, this system can be modelled as a series of resistors (Figure 8). Each resistor represents a different medium in which heat travels through. The current represents the heat transfer rate, q . Applying Ohm's Law (Equation 3) and solving for current results in the heat transfer rate of the system (Equation 4).

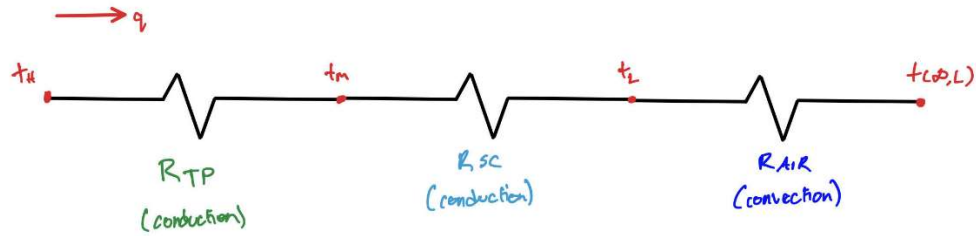


Figure 8 - Thermal Resistance Network

$$V = IR$$

Equation 3 - Ohm's Law

Where:

- V = voltage [V]
- I = current [A]
- R = resistance [Ω]

$$I = \frac{V}{R}$$

$$q = \frac{\Delta T}{R_{eff}}$$

$$q = \frac{T_H - T(\infty, L)}{R_{TP} + R_{SC} + R_{AIR}}$$

$$q = \frac{T_H - T(\infty, L)}{\frac{l_2}{k_2 A_2} + \frac{l_1}{k_1 A_1} + \frac{1}{h_L A_3}}$$

Equation 4 - Heat Transfer Rate of Enclosure Derived from Thermal Resistance Method

Where:

- q = heat transfer rate [W]
- T_H = SSD temperature [K]
- $T(\infty, L)$ = ambient temperature [K]
- k_i = thermal conductivity [$\frac{W}{mK}$] (conduction)
- h_L = thermal conductivity [$\frac{W}{m^2K}$] (convection)
- A_i = surface area [m^2]

Thus, to increase the performance of the device the heat transfer rate must be maximized. The formula for heat transfer (Equation 4) contains variables that can be altered to increase q . Theoretically the difference in temperature could be increased by increasing T_H and/or lowering $T(\infty, L)$. Evidently however this is not feasible as increasing the SSD temperature would cause it to begin thermal throttling, reducing performance. Lowering the outside temperature cannot be done consistently.

Instead, the thickness of the thermal pad and shell cover could be decreased in theory. Again, this method also lacks practicality as reducing the thickness of the thermal pad could result in less contact between the two surfaces (air pockets) which increases the effective resistance (lower heat transfer rate). Reducing the thickness of the shell cover would affect the enclosure's structural integrity, diminishing the overall ruggedness.

Therefore, the only practical alterations that can be made are to the thermal conductivity constants (k_i) and the surface area (A_i).

3.2.2 Geometry

The first possible modification is to alter the geometry of the shell. The objective is to increase the exterior surface area as much as possible to aid in the convective heat transfer. To keep costs low, initial prototyping and testing is done using CFD (Computational Fluid Dynamics) modelling.

The first step is to build a 3D model using CAD (Computer Aided Design) based on basic dimensions and photos of the Trebleet enclosure (Figure 9).

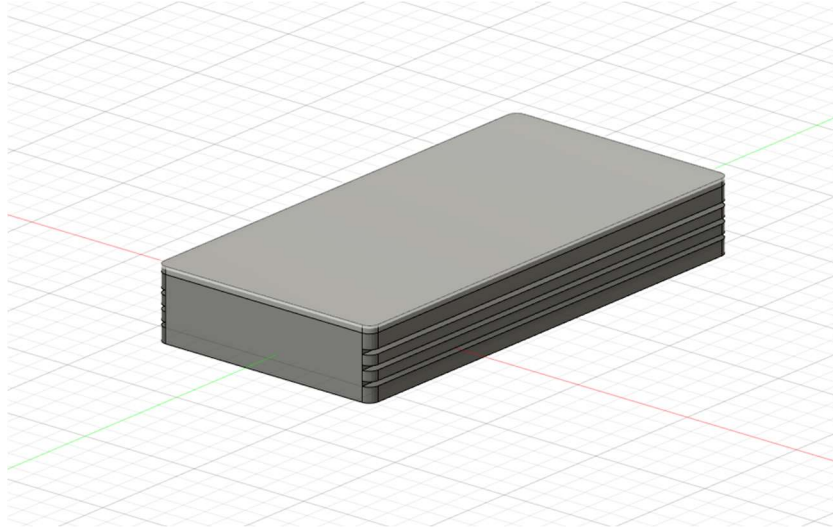


Figure 9 - CAD Model of Trebleet Enclosure

To increase the likelihood of convergence in the CFD simulation, the model must be simplified. Omitting fine details such as fasteners, stand-offs, and paint reduce the model's complexity with negligible affects to accuracy. Therefore, the SSD is modeled as a single rectangular volume based on the listed 980 Pro dimensions to further simplify the simulation. Also, the circuit board is omitted as the majority of heat generation comes from the SSD. Finally, the model is cut in half along its length axis. Because the enclosure is effectively symmetrical along this axis it is possible to remove half of the model to simplify the simulation (Figure 10).

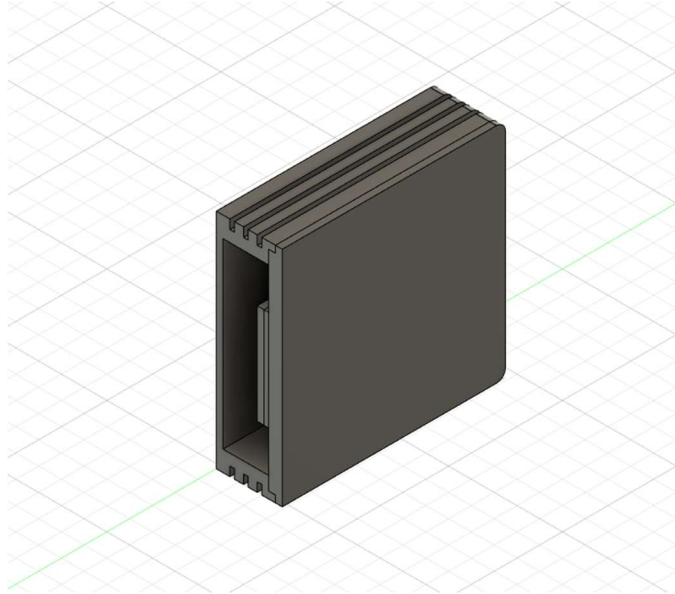


Figure 10 - Split CAD Model of Trebleet Enclosure

Several prototype designs were modelled which increase exterior surface area while maintaining structural integrity and compact size. Different shaped fins were placed on the top surface of the shell cover in a grid pattern (increasing the height an additional 5 mm) (Figure 11).

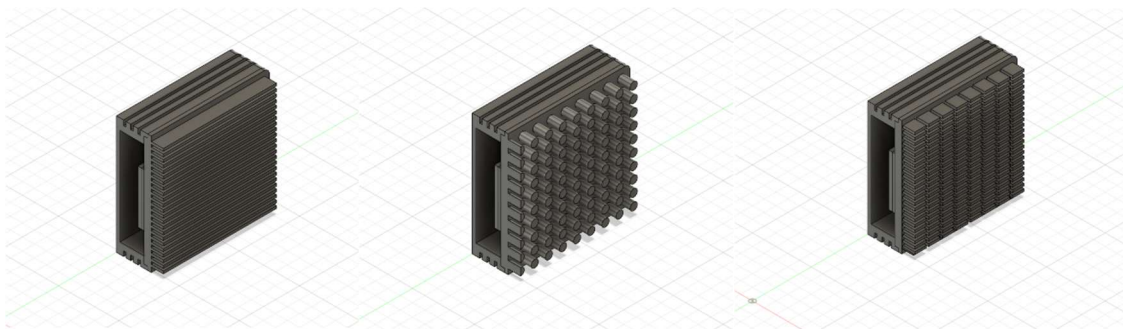


Figure 11 - Proposed Fin Designs (left - plate fin, middle - cylindrical fin, right - pin fin)

3.2.3 Material

The second possible modification is to change the material used in the shell. Materials have varying thermal conductivities (k). By choosing a structural material with increased heat conductivity the enclosure will move heat out faster.

Simulations were conducted comparing the factory Aluminum shell ($k = 236 \frac{W}{mK}$) versus Copper ($k = 401 \frac{W}{mK}$) [19].

3.2.4 CFD Simulations

To compare the cooling performance of the prototypes, Simscale's Conjugate Heat Transfer v2.0 algorithm was used. This algorithm is designed to analyze heat transfer between solid and fluid domains [20]. It is able to model both conduction and convection modes simultaneously.

3.2.4.1 Defining Domains

Following Simscale's recommendations, the exterior fluid boundary was created according to Figure 12. This rule of thumb should be sufficient in encapsulating the convection boundary. An interior fluid boundary was then created in the enclosure. Both fluid regions were modelled as air.

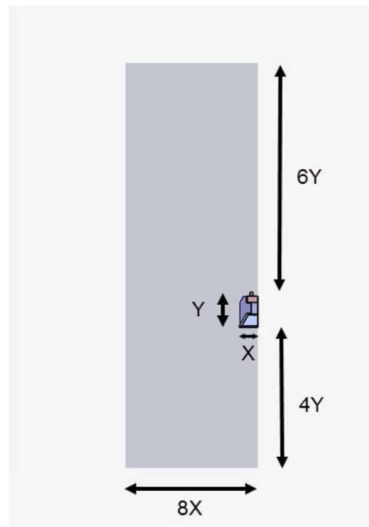


Figure 12 - Exterior Fluid Boundary Rule of Thumb [21]

The solid bodies were modelled with the following materials:

- Shell cover: Aluminum or Copper depending on the trial.
- Shell box: Aluminum.

- Thermal pad: Silicone.
- SSD: Silicon

Although the SSD contains several different elements in reality, modelling the device as Silicon simplifies the simulation. This simplification should have minimal effect on the results as the majority of heat generation comes from the memory modules which are mostly Silicon.

All material properties were taken at STP (Standard Temperature and Pressure).

3.2.4.2 Initial Conditions

To simulate some air flow across the enclosure, a cylindrical sub-domain was created fully encapsulating the enclosure along its length axis. A speed of 0.05 m/s was assigned to the cylindrical domain as shown in the figure below (Figure 13).

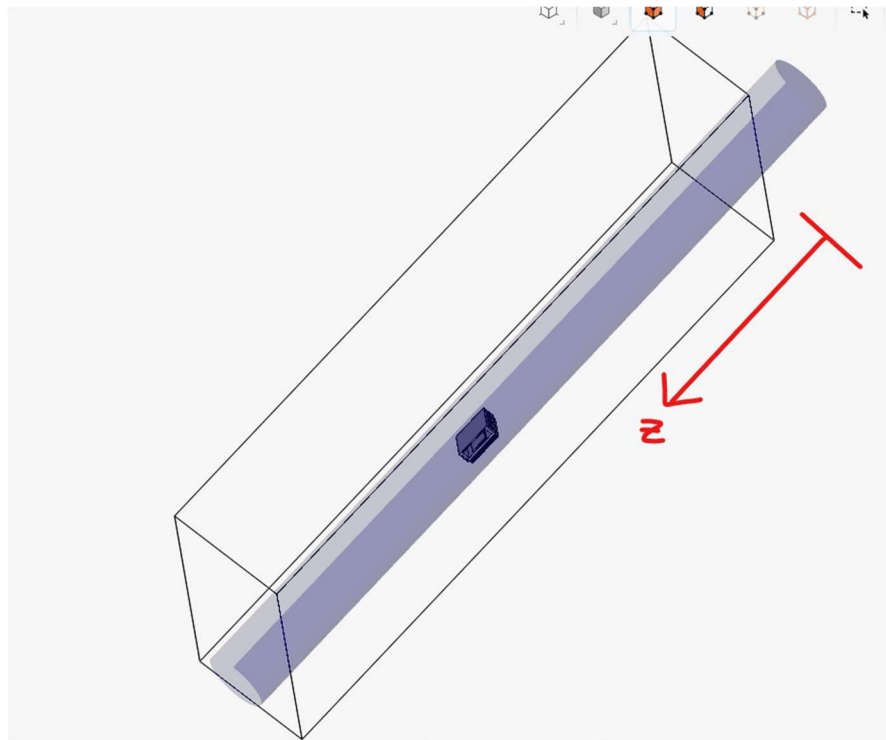


Figure 13 - Initial Air Speed Domain

The SSD was set at 35 degrees Celsius to simulate normal idle operating temperatures. It was also set as a power source to model the heat generated. Samsung claims the 980 Pro expels a maximum of around 9 Watts in the form of heat during peak operation [10]. Because the simulation only model's half of the enclosure, the SSD was set to 4.5 Watts of heat generation.

3.2.4.3 Boundary Conditions

All faces of the exterior domain excluding the front face were modelled as natural convection inlets/outlets. This setting allows air to enter and exit to the surrounding atmosphere freely. The ambient temperature was set at room temperature (19.85 degrees Celsius).

The front face of the exterior domain, interior domain, shell, thermal pad, and SSD were all modelled as symmetrical. Rather than modelling the entire device, this setting allows the user to exclude symmetrical geometries. This reduces the computing power necessary with no effect to the final result.

3.2.3 Results

The resulting cooling performance of the factory design and prototypes was mostly consistent with expectations. The average temperature of the SSD of the respective enclosure designs can be seen in the figure below (Figure 14). The plot shows the SSD begins to approach steady-state equilibrium with its surroundings after around 150 seconds of peak operation. The equilibrium temperatures from highest to lowest are as follows:

1. Factory cover ($\sim 100^{\circ}\text{C}$)
2. Pin fin cover ($\sim 95^{\circ}\text{C}$)
3. Cylindrical fin cover ($\sim 90^{\circ}\text{C}$)
4. Plate fin cover ($\sim 85^{\circ}\text{C}$)

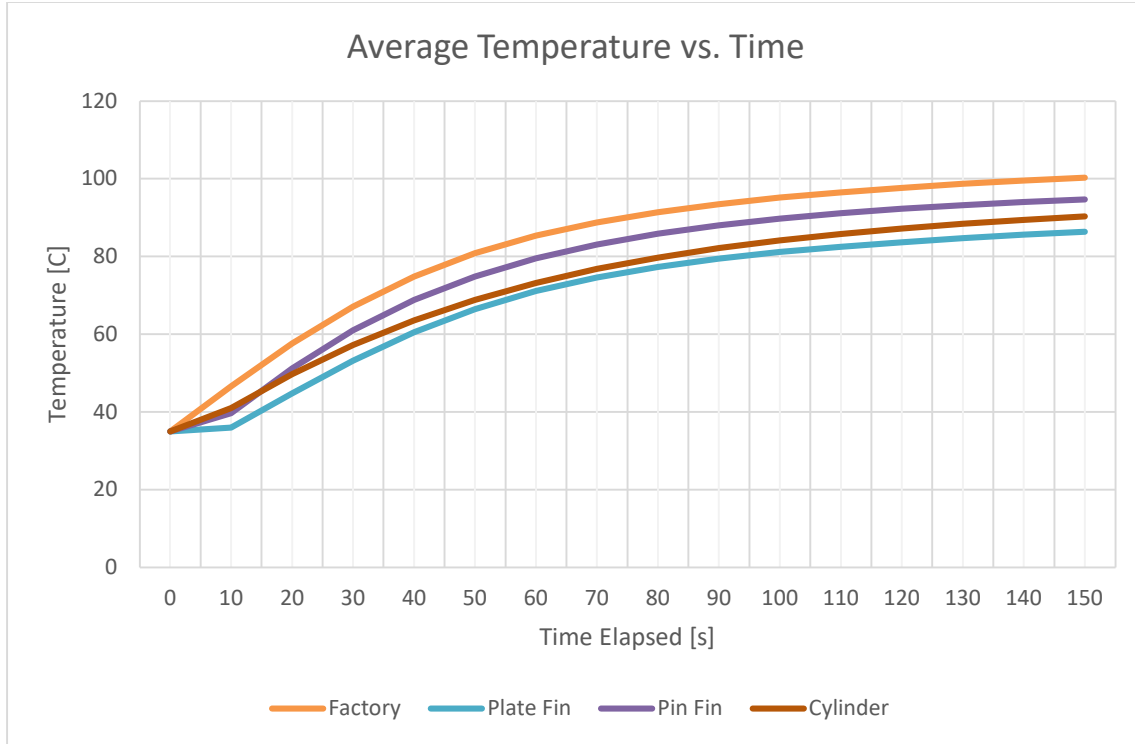


Figure 14 – Average Temperature of SSD

The plate fin design performs the best as it has the lowest equilibrium temperature when compared to the rest. The plate fin also displays lower temperatures throughout the simulation. Although the SSD is unlikely to operate at peak power over an extended period like this, these results indicate the plate fin design will keep the SSD the coolest.

Plotting the rate of change of the average temperature of the SSD further demonstrates the superiority of the plate fin design. As shown below in Figure 15, the plate fin design displays the lowest rate of temperature change before reaching equilibrium. This indicates the plate fin design has the highest heat transfer rate.

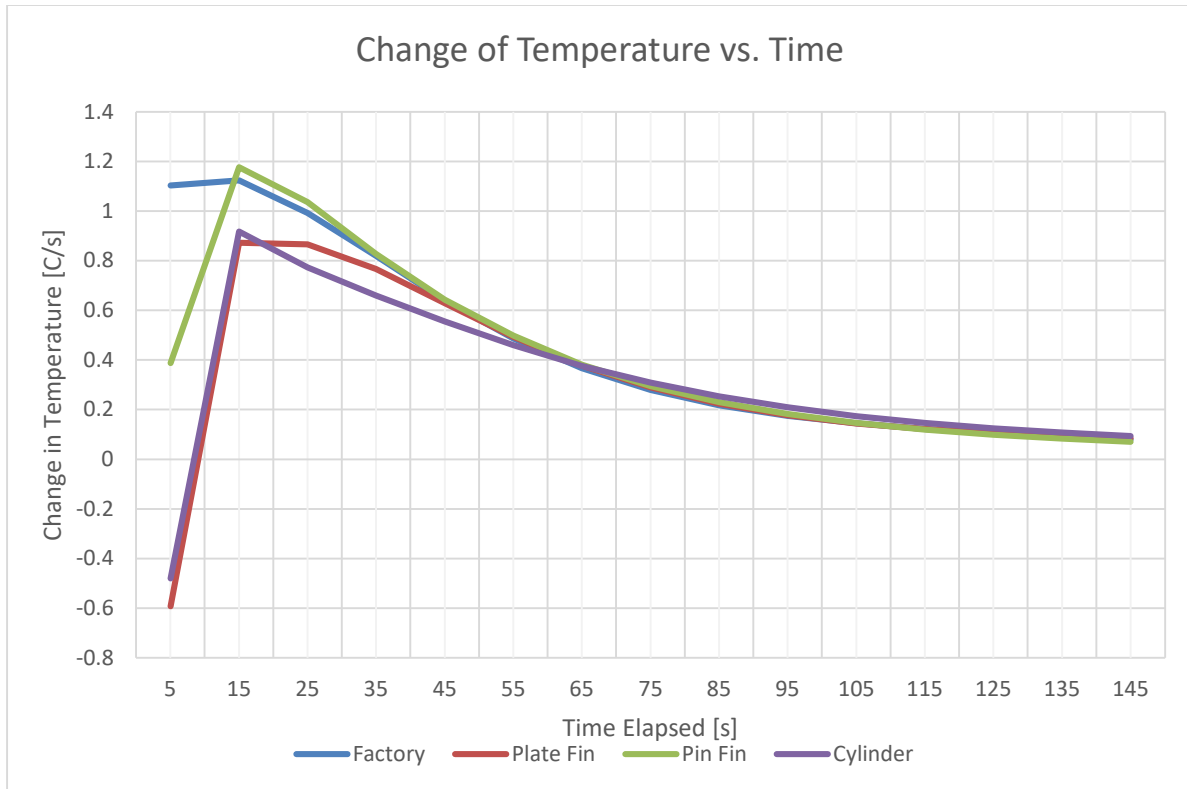


Figure 15 - Rate of Average SSD Temperature Change

Because the plate fin design performs the best given its geometry, increasing its conductivity should further improve its cooling capability. Unfortunately, numerical data was not available for the Copper plate fin simulation as the Simscale trial period ended prematurely. Regardless, comparing the temperatures at 10, 30, and 50 seconds respectively provides some insight into the difference in cooling performance between Copper and Aluminum (Figure 16).

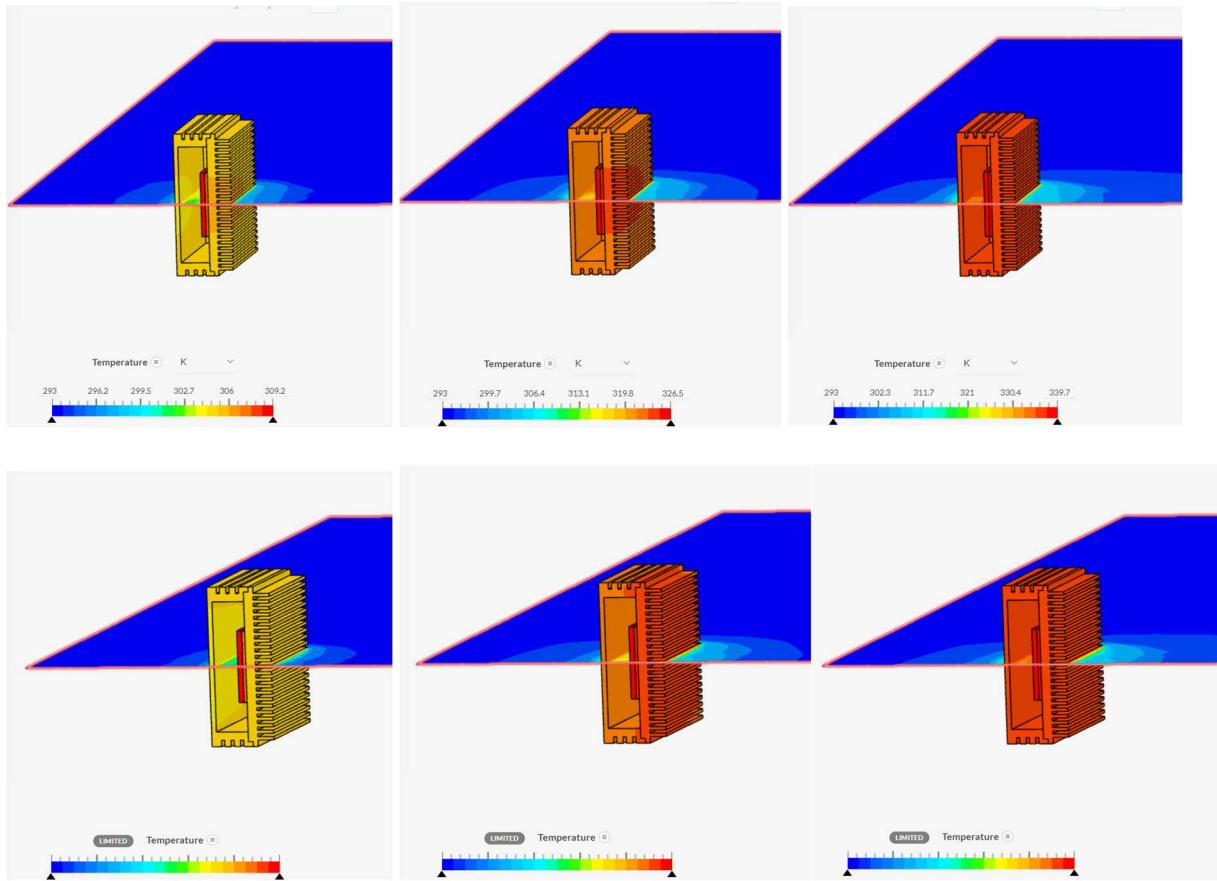


Figure 16 - Temperature of Aluminum (top) and Copper (bottom) at 10, 30, and 50 s (left to right)

Although the exact temperatures of the SSD in the Copper enclosure are unknown, visually comparing the temperature of the two designs reveals the Copper shell's temperature rises faster than the Aluminum shell. Considering both simulations share the same initial and boundary conditions this indicates the Copper design transfers heat from the SSD faster than Aluminum, corresponding with expectations.

Thus, the optimal enclosure design incorporates both plate fins on the shell cover as well as Copper instead of Aluminum (Figure 17).

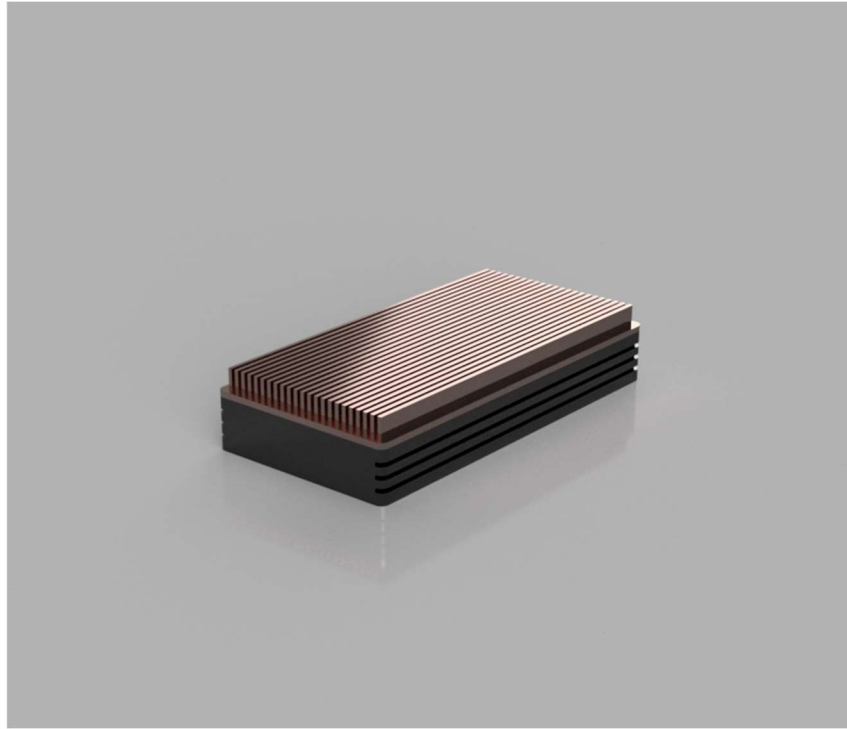


Figure 17 - Render of Final Enclosure Design

3.3 Remaining Challenges

The initial CFD simulations provide a good insight into the expected cooling performance of the enclosure. They indicate the copper plate fin design dispels heat with the highest efficiency. However, these results have only been shown to hold true with a specific set of initial and boundary conditions. To further validate these claims more simulations should be conducted with varying parameters.

4.0 Conclusions

4.1 Conclusions

A custom external SSD enclosure was designed in response to the need of a reliable, fast, and compact external storage device which stores a minimum of 1 TB. The enclosure was designed to operate without internet connectivity while maximizing read/write speeds, cooling performance, and minimizing construction costs.

The proposed design incorporates a Samsung 980 Pro 1 TB SSD because of its exceptionally fast sustained transfer speeds and relatively low price. The 980 Pro is housed in a Trebleet 2 in 1 TB3 enclosure which supports both USB 3.1/3.2 and TB3 simultaneously in a small package.

Finally, additional modifications to the Trebleet enclosure were designed to further improve the transfer speeds and heat dissipation efficacy of the new device. Plate fins were added to the enclosure's cover to increase the exterior surface area and the legacy aluminum was replaced with copper to increase the conductivity. These changes were modelled with CFD simulations to ensure their effectiveness when compared to the factory design.

4.2 Next Steps / Recommendations

After finalizing the design of the new external storage device, the next critical step is to assemble the required parts and materials to build the enclosure. While the majority of the parts can be purchased directly, the shell cover requires custom work. Using the standard factory cover from the Trebleet enclosure as a template, precise dimensional measurements will be required to re-model the final shell cover in CAD. The final CAD model must then be submitted to a machine shop to be manufactured.

The storage device can then finally be fully assembled. Once assembled the last step is to test the device to ensure it meets all the constraints and criteria in practice.

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